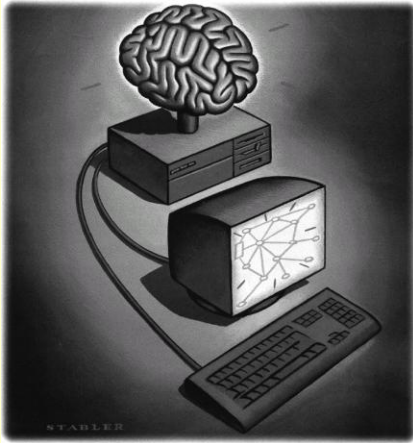




# NN Parameters (Practise)

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Department of Information and Computer Engineering  
Politeknik Elektronika Negeri Surabaya

# Multi Layer Perceptron

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.neural_network import MLPClassifier as MLP

dataset = pd.read_csv('XOR.csv')
data = dataset.iloc[:,0:-1]
label = dataset.iloc[:,-1]

iter_limit = range(300)
iter = [i+1 for i in iter_limit]
iter_num = np.array(iter)

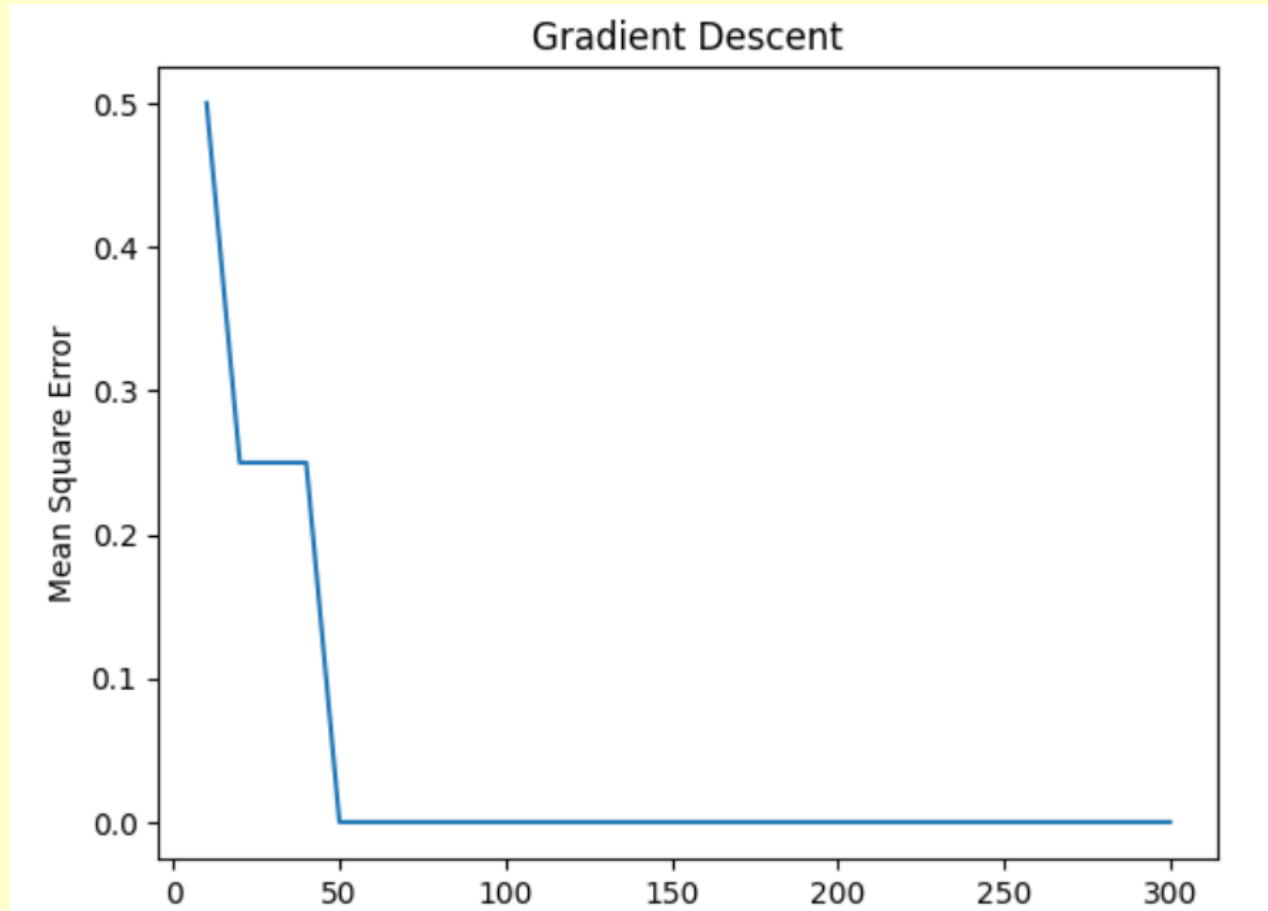
def mse(actual, pred):
    actual, pred = np.array(actual), np.array(pred)
    return np.square(np.subtract(actual,pred)).mean()
```

# Multi Layer Perceptron

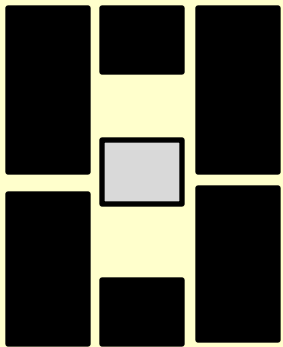
```
for i in range (0,iter_num.size):
    model = MLP(hidden_layer_sizes=(7), max_iter=iter[i],
activation = 'logistic', learning_rate_init=0.1)
    model.fit(data,label)
    pred=model.predict(data)
    if i==0 :
        error = np.array(mse(label, pred))
    else :
        error = np.append(error,mse(label, pred))
    print('score:', model.score(data,label))

plt.plot(iter,error)
plt.xlabel('Iteration No.')
plt.ylabel('Mean Square Error')
plt.title('Gradient Descent')
plt.show()
```

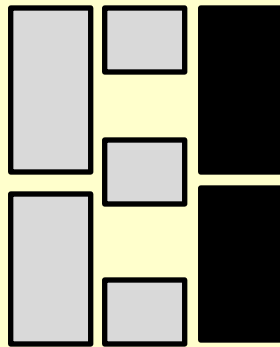
# Output



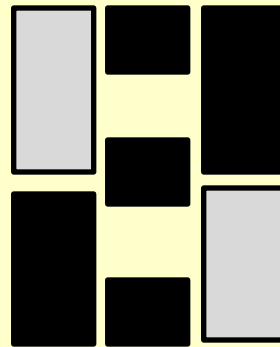
# 7 Segments



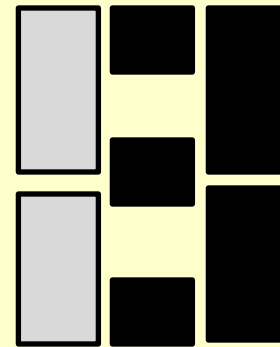
0



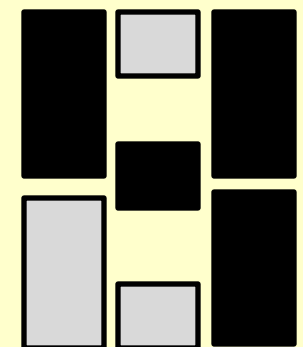
1



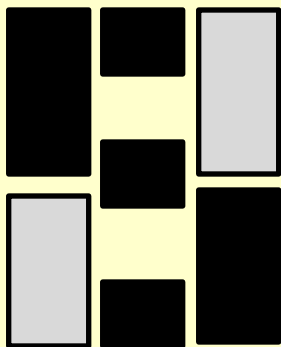
2



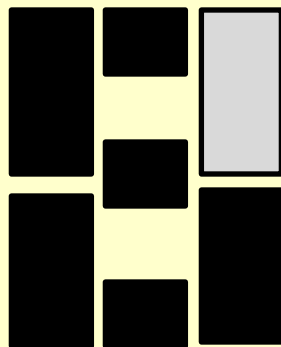
3



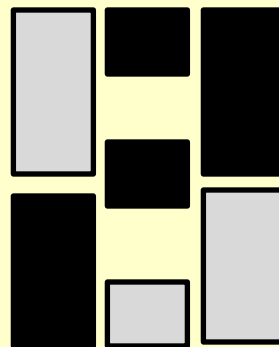
4



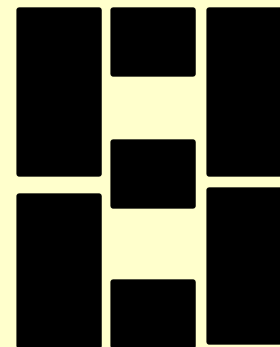
5



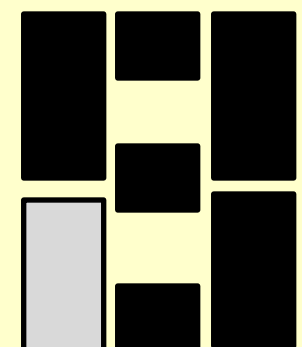
6



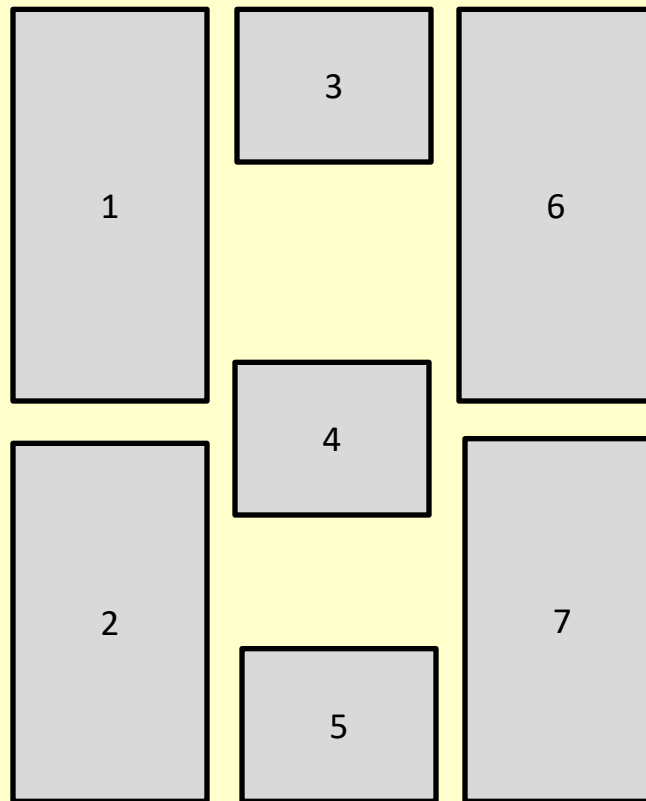
7



8



9

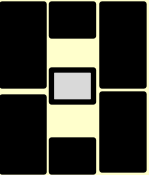
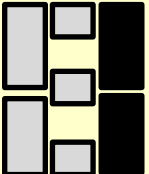
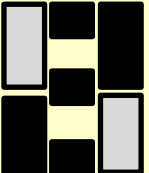
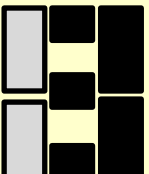
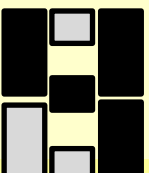


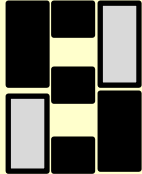
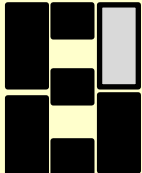
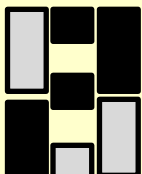
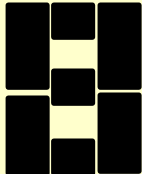
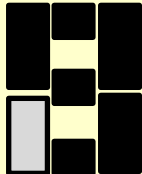
# Input Sequence

|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 2 | 0 | 1 | 1 | 1 | 1 | 1 | 0 |
| 3 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 4 | 1 | 0 | 0 | 1 | 0 | 1 | 1 |

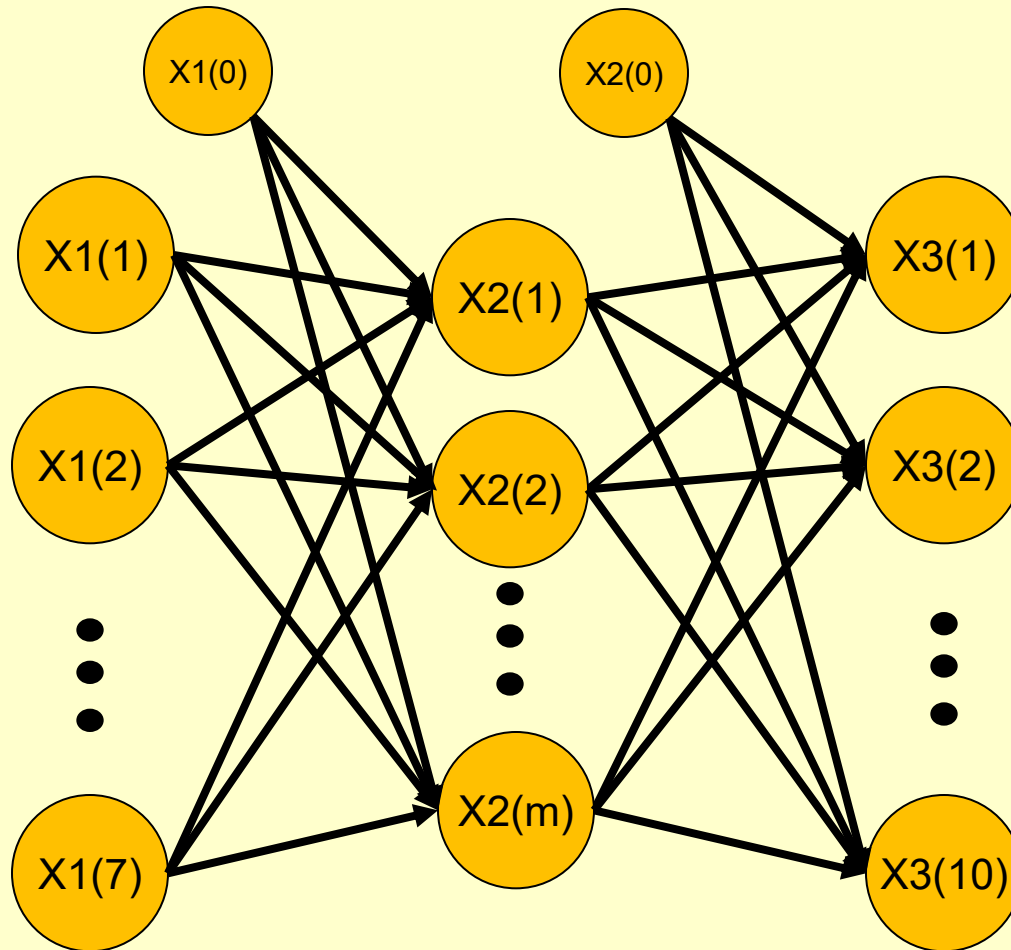
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| 5 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 6 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| 7 | 0 | 1 | 1 | 1 | 0 | 1 | 0 |
| 8 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 9 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |

# Target

|   | 0                                                                                   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|-------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|
| 0 |    | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 |    | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 |    | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 |   | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 |  | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |

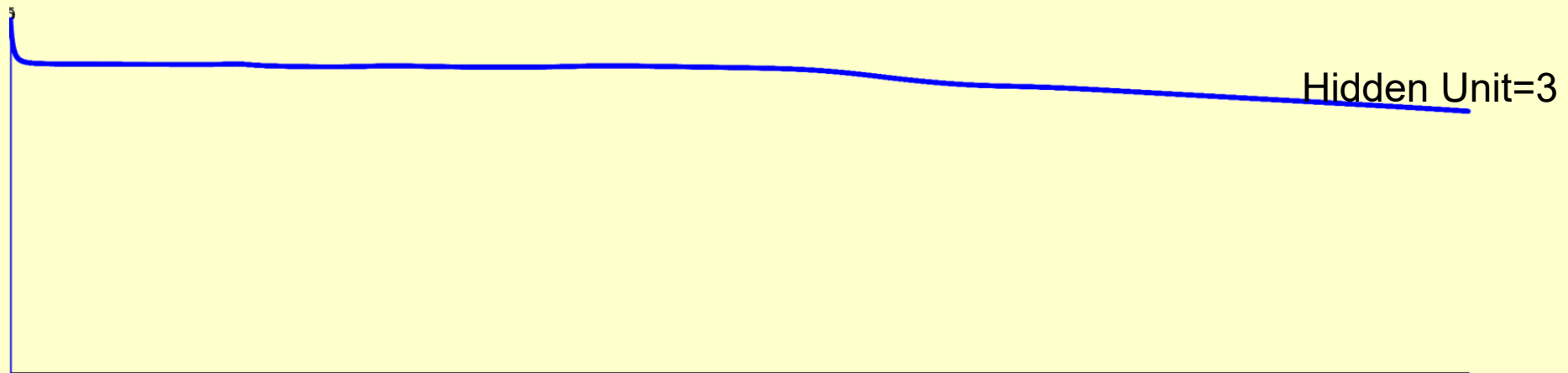
|   | 0                                                                                    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|--------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|
| 5 |    | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 |    | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 7 |    | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 8 |   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 9 |  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

# 7-Segment Model



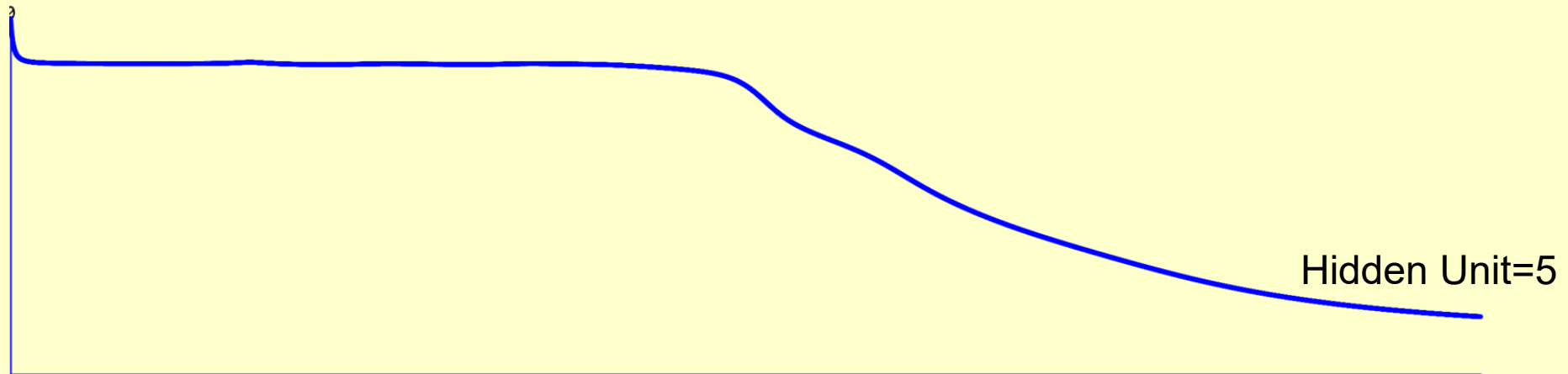
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



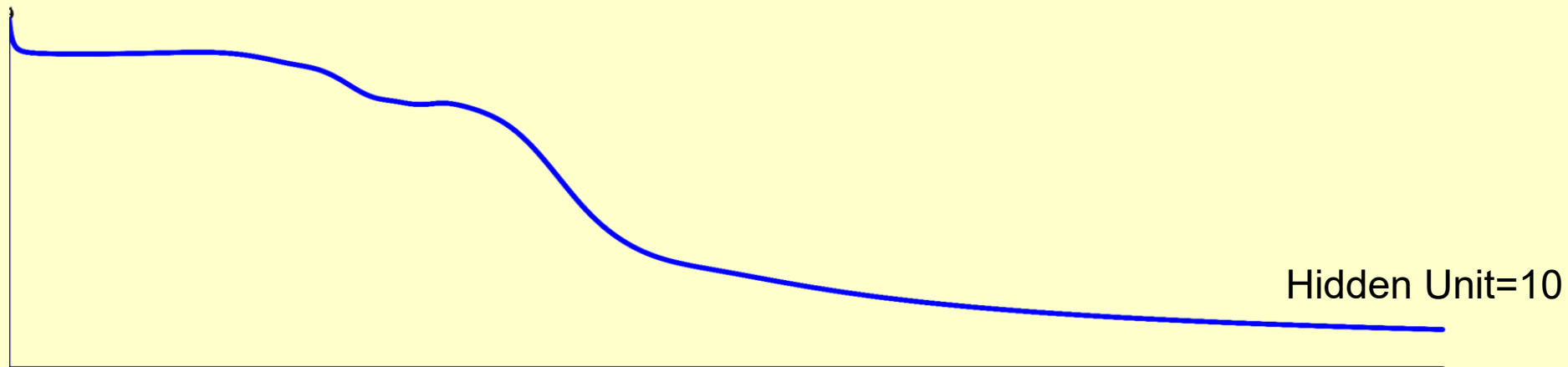
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



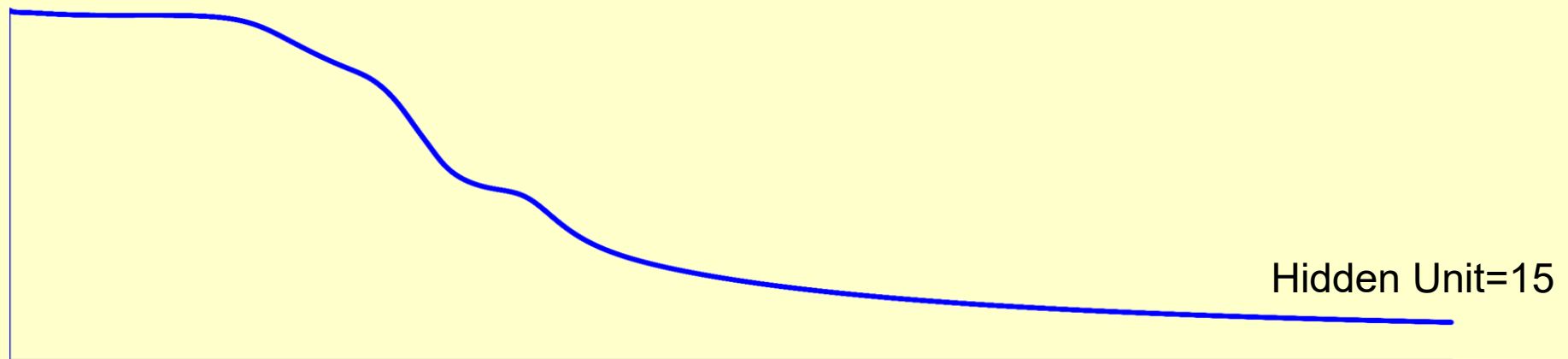
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



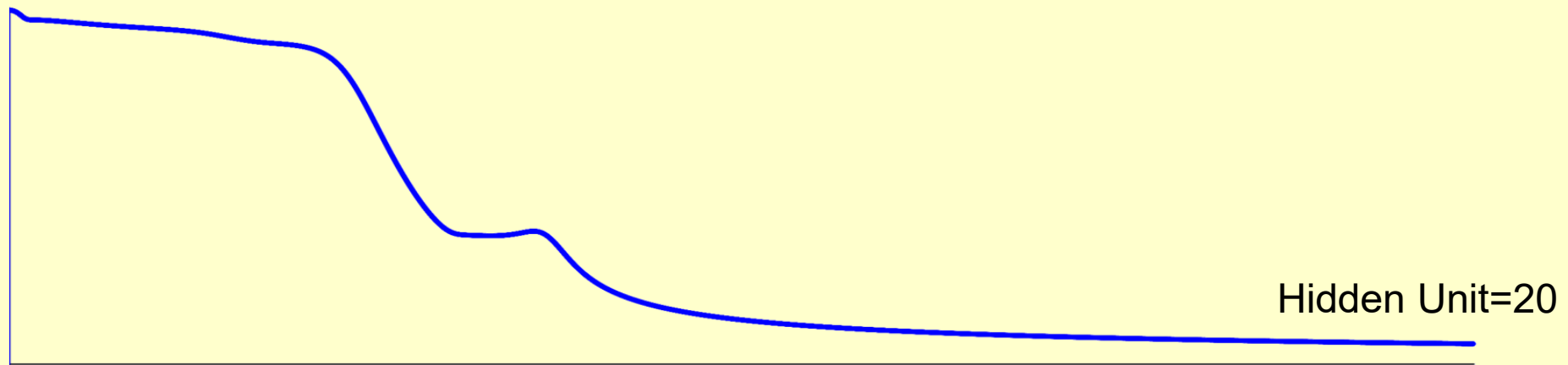
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



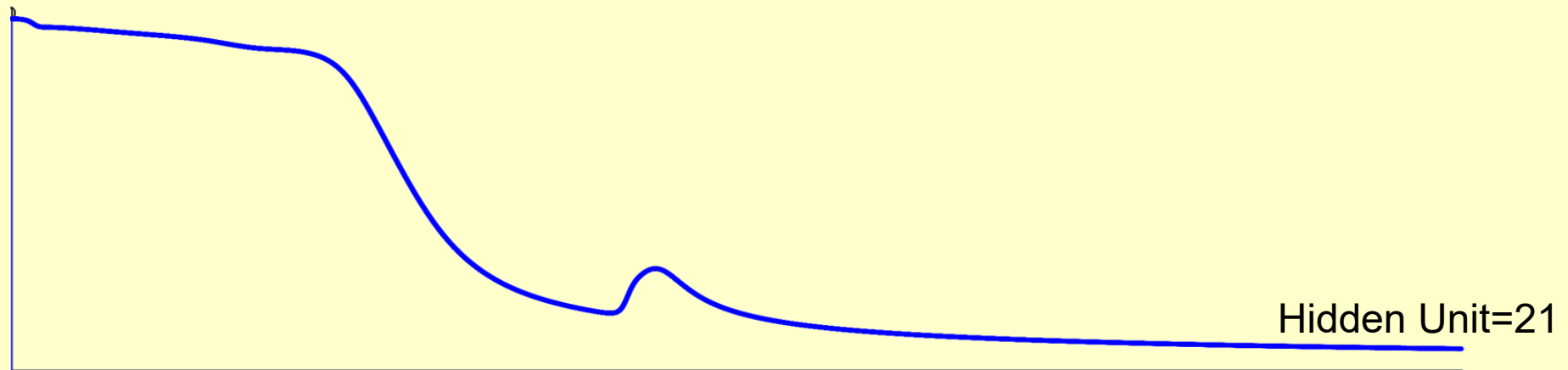
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



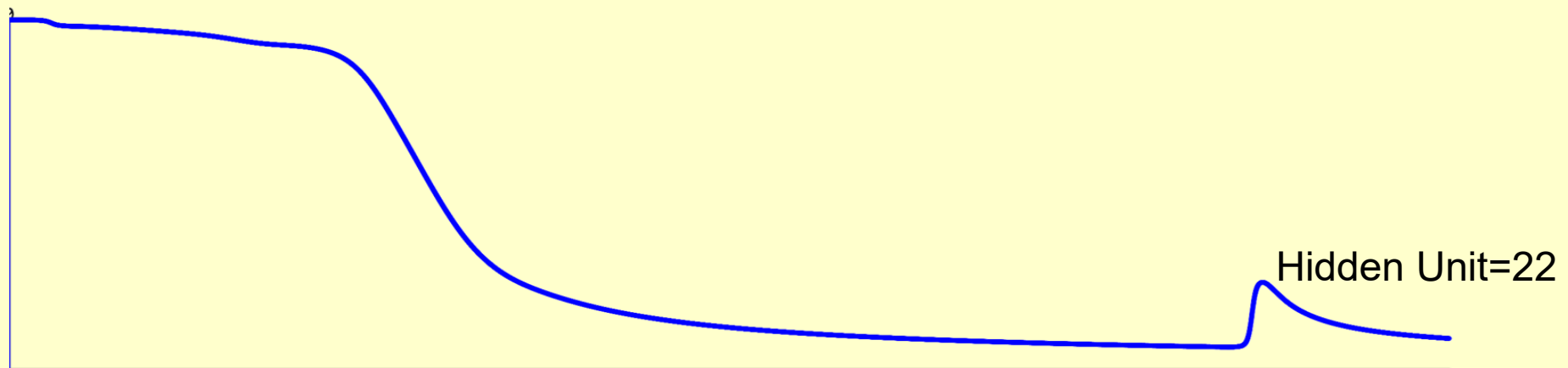
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



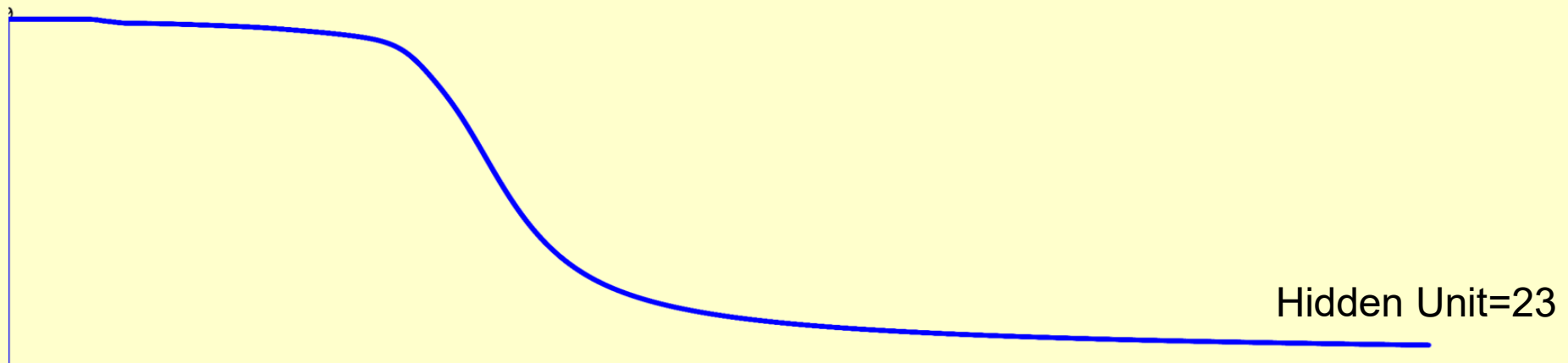
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )



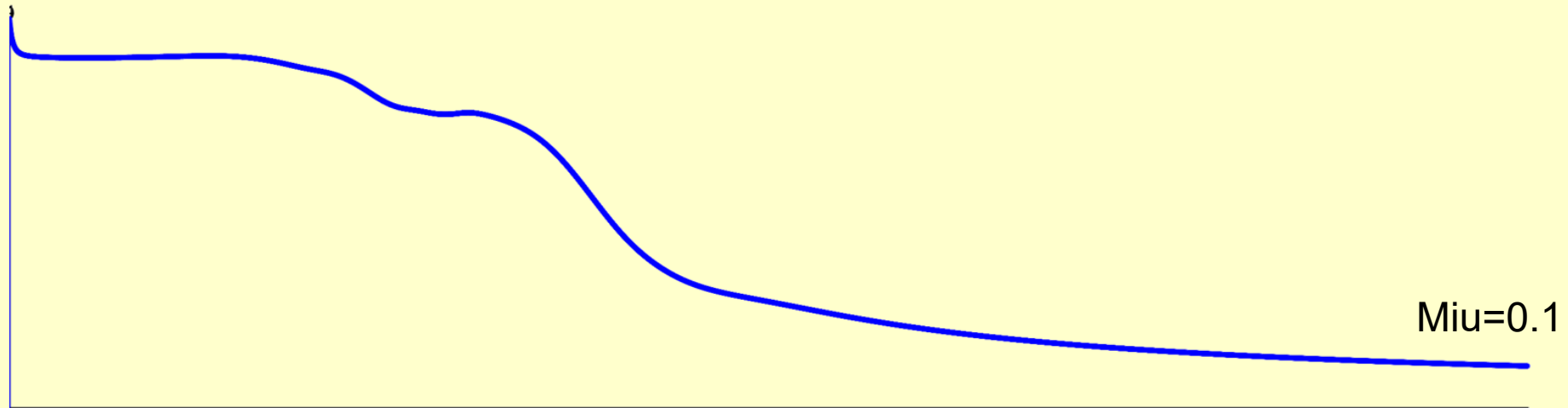
# Pengaruh Hidden Unit terhadap SSE

7-Segment Classification  
( $\mu=0.1$ )

Hidden Unit=24

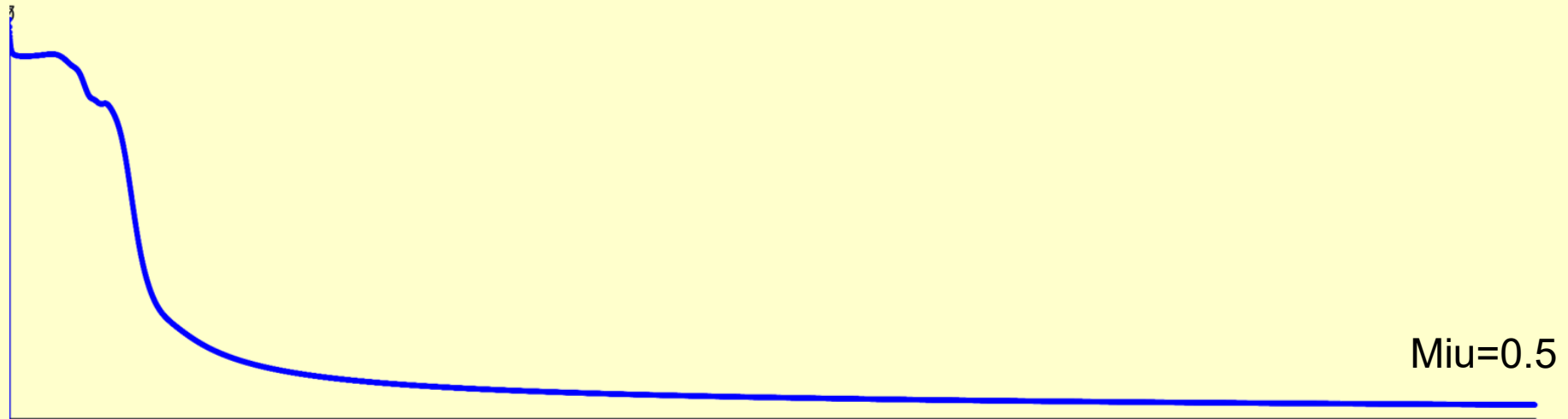
# Pengaruh Learning Rate terhadap SSE

7-Segment Classification  
(Hidden Unit=10)



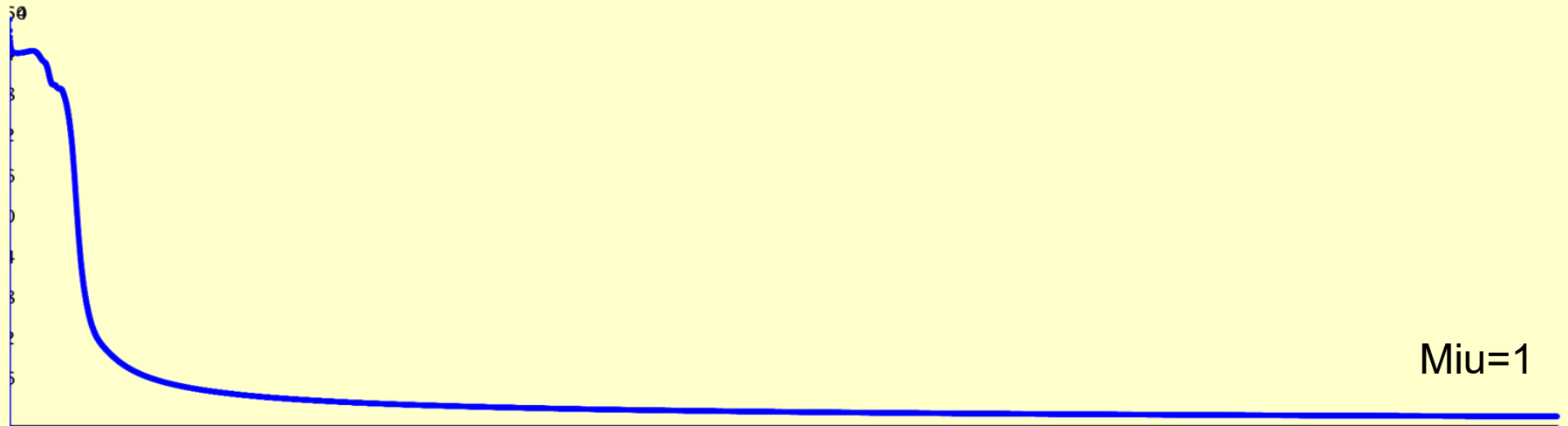
# Pengaruh Learning Rate terhadap SSE

7-Segment Classification  
(Hidden Unit=10)



# Pengaruh Learning Rate terhadap SSE

7-Segment Classification  
(Hidden Unit=10)



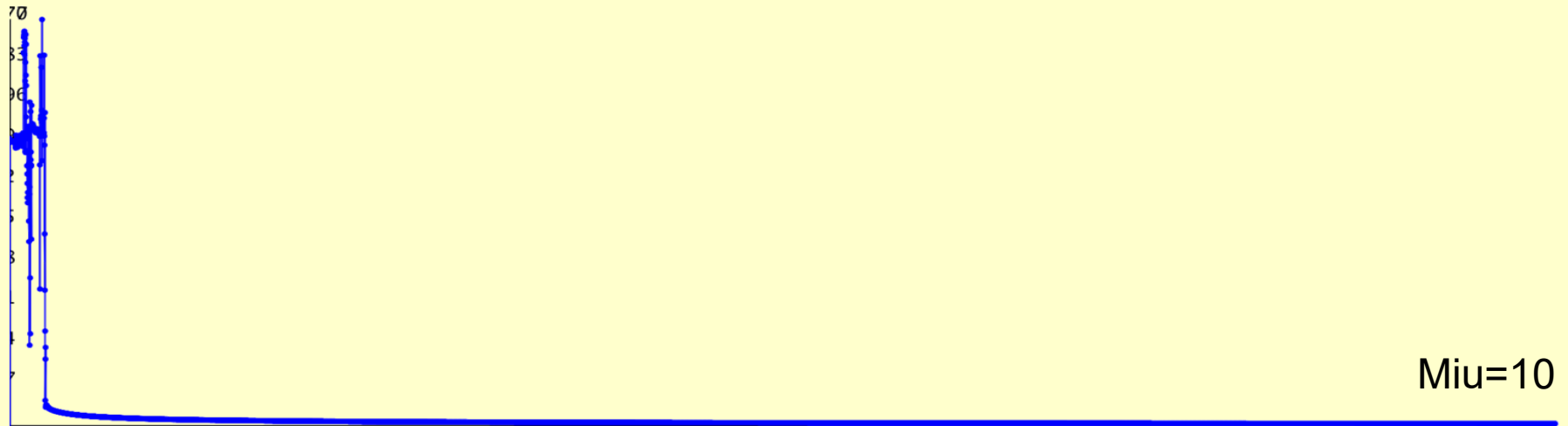
# Pengaruh Learning Rate terhadap SSE

7-Segment Classification  
(Hidden Unit=10)



# Pengaruh Learning Rate terhadap SSE

7-Segment Classification  
(Hidden Unit=10)



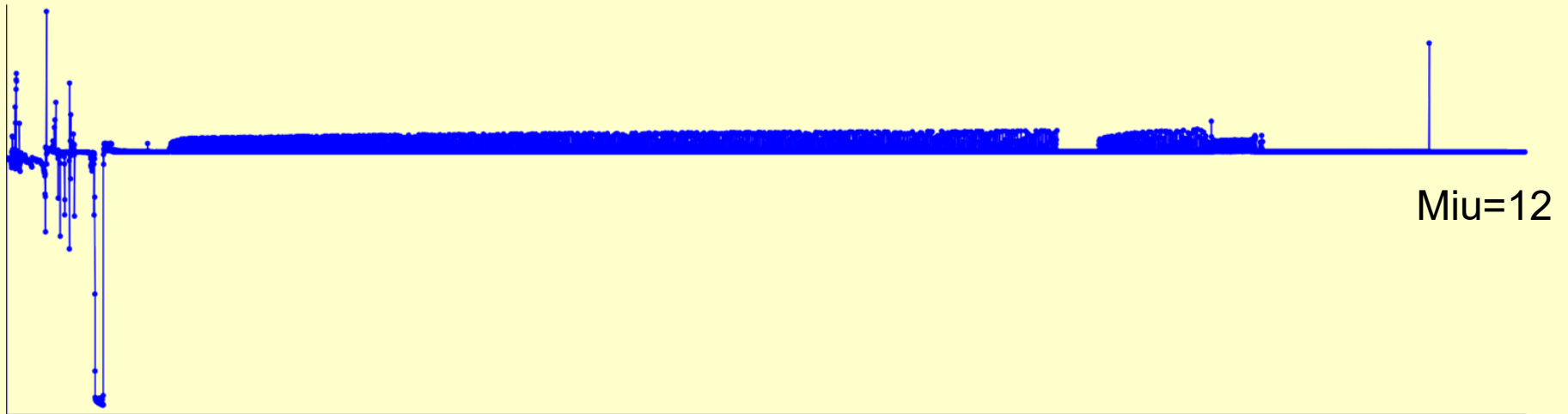
# Pengaruh Learning Rate terhadap SSE

7-Segment Classification  
(Hidden Unit=10)



# Pengaruh Learning Rate terhadap SSE

7-Segment Classification  
(Hidden Unit=10)



# Assignment #4

1. Lakukan Multi Perceptron untuk data XOR dan analisa parameter untuk Fungsi Aktivasi & jumlah Hidden Layer dengan menggambarkan Gradient Descentnya seperti di atas. Carilah parameter yang tepat hingga mendapatkan error minimal.
2. Buatlah data training untuk 7-Segment. Lakukan yang sama seperti no 1 untuk  $\mu=0.1$ , layer =3,5,10,15,20,21,22,23,24
3. Gambarkan juga Gradient Descent untuk jumlah layer 10 dengan  $\mu = 0.1, 0.5, 5, 10, 11, 12$

# Pengumpulan Tugas

- Buatlah coding dengan Bahasa pemrograman/tools apapun untuk semua assignment
- Buatlah laporan dalam slide ppt. Laporan terdiri dari screenshot coding dan hasil running untuk setiap assignment.
- Simpan laporan dalam file pdf dengan format penamaan:  
KKIT\_M4\_NRP\_namadepan.pdf
- Upload file tersebut ke ETHOL
- Deadline upload:
  - Reguler & RPL kelas A, B : Selasa, 22 September pk 23.00